

Wei Cheng Chiu

AI Solutions Architecture | Generative AI / LLM Systems | NVIDIA GPU Computing

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SUMMARY

Field Application Engineer pursuing customer-facing **AI Solutions Architect** roles, with hands-on experience translating requirements into production-ready, on-premises **Generative AI (GenAI) / LLM** deployments on NVIDIA GPU platforms. Work spans proof of concept (PoC), Linux / Kubernetes operations, model deployment and evaluation, CUDA / TensorRT-LLM / vLLM / Triton inference, performance optimization, troubleshooting, acceptance, and handover. At Taiwan AILabs, I put on-premises LLM systems (FedGPT) into customer environments. Previously **sole developer** of an enterprise multi-agent AI platform on DGX-class hardware, taken from PoC to production at **two enterprise customers**. Active upstream — **17 merged/landed patches** across FlashInfer, vLLM, SGLang, PyTorch, Dynamo, CUTLASS, TensorRT-LLM and LMDeploy, concentrated on early consumer-Blackwell (**SM120**) and **NVFP4** enablement. Research background in AI/LLM security and privacy-preserving ML.

TECHNICAL SKILLS

LLM Serving & GPU	vLLM, TensorRT-LLM, SGLang, Triton, NVIDIA NIM, Dynamo, FlashInfer, NCCL, FlashAttention, KV-cache, Quantization (FP8 / INT4 / NVFP4); CUDA C/C++, WMMA & <code>mma.sync</code> Tensor Cores, CUTLASS, Nsight Systems & Compute
GenAI Apps & Agents	Generative AI, Deep Learning, Model Fine-Tuning, Deployment & Evaluation, Multi-Agent Orchestration (LangChain, LangGraph), Hybrid RAG (Vector + Keyword + Knowledge Graph), LLM Guardrails, Function Calling / Tool Use, MCP, Coding Agents
Infrastructure & Ops	Linux, Kubernetes, Docker, On-prem GPU Serving, Distributed Tracing; H100, DGX Spark (GB10), Blackwell workstation hardware
Privacy & Security ML	Federated Learning, Differential Privacy, Secure Multi-Party Computation (MP-SPDZ), Safety Alignment
Programming & Frameworks	Python, TypeScript, Rust, C/C++; PyTorch, FastAPI, NestJS; PostgreSQL, Milvus, Redis

EXPERIENCE

Taiwan AILabs <i>Field Application Engineer</i>	Jul 2026 – Present <i>Taipei, Taiwan</i>
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- Deploy production-ready, on-premises LLM systems (**FedGPT**) into customer environments; translate customer requirements and site constraints into Linux / Kubernetes / GPU-serving plans, preflight checks, and acceptance criteria
- Troubleshoot cross-layer failures spanning containers, GPU runtime, storage, networking, certificates, and model-serving endpoints; explain performance, reliability, and operational trade-offs through handover

SYNCRBOTIC <i>Machine Learning Engineer — Multi-Agent AI, RAG, LLM Serving</i>	Sep 2025 – Jun 2026 <i>New Taipei City, Taiwan</i>
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- **Sole developer** of an enterprise multi-agent AI platform on DGX-class hardware — owned end-to-end from PoC through production and rolled out at **two enterprise customers**, partnering with each through deployment and troubleshooting
- Scoped ambiguous enterprise requirements into a working multi-agent system and **unified local and commercial LLM backends** under one routing layer, removing per-team model wrappers
- Designed a planning–execution–validation orchestration with topological scheduling and parallel agent dispatch for multi-step workflows
- Built a hybrid retrieval layer on LightRAG (vector + keyword + knowledge graph) with embedding-based semantic caching, cutting redundant LLM calls and keeping answers consistent on repeat queries
- Owned production reliability — prompt/pipeline regression tests, output validation across quality and safety, distributed tracing; added CUDA-accelerated hot paths on critical inference

Advantech <i>AI Agent Engineer (Intern)</i>	Jun 2025 – Aug 2025 <i>Taipei, Taiwan</i>
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- Built coding agents for internal developer tooling using **function calling / tool use** — automating code review, test generation, and static analysis across the SDLC; benchmarked tool-use reliability across commercial and local LLM backends and designed reusable prompt templates adopted by the team

TSMC (Taiwan Semiconductor Manufacturing Co.) <i>R&D Alternative Military Service — Facility Engineer</i>	May 2021 – Oct 2022 <i>Tainan, Taiwan</i>
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- Supported R&D fab facility operations at the world’s leading semiconductor foundry under the national R&D alternative military service program

OPEN SOURCE

SM120 / NVFP4 enablement across the LLM-inference stack
FlashInfer · vLLM · SGLang · PyTorch · Dynamo · CUTLASS · TensorRT-LLM · LMDeploy

17 merged/landed
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- Kernel-level support for consumer Blackwell (**SM120**) and **NVFP4** carried upward through kernels → engines → disaggregated serving
- Focus on *silent-correctness* bugs — the ones where tests pass and the answers are still wrong

RESEARCH & PUBLICATIONS

SelGrad: Selective Gradient Projection for Efficient Safety Alignment Against Harmful Fine-Tuning — first author, with Shao-Jui Wang. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, under review, 2026. Preserves LLM safety alignment under harmful/adversarial fine-tuning at lower compute than full re-alignment.

Research at NTUST (advised by Prof. Shao-Jui Wang) in model fine-tuning, experiment design, and evaluation for **AI/LLM security** and **privacy-enhancing ML** — federated learning, differential privacy, and secure multi-party computation.

EDUCATION

National Taiwan University of Science and Technology (NTUST)
M.S. in Computer Science and Information Engineering — GPA 4.09

Aug 2024 – Apr 2026
Taipei, Taiwan

National Taiwan University (NTU)
M.S. in Structural Engineering (2020) | B.S. in Civil Engineering (2018)

Sep 2014 – Jun 2020
Taipei, Taiwan

CERTIFICATIONS & AWARDS

NVIDIA DLI — Accelerated Computing with CUDA C/C++ and CUDA Python (5 certificates) · NICS Privacy-Enhancing Technologies (PETs) Competition — Rank 3/11 (DP synthetic data via MP-SPDZ) · International ICT Innovative Services Awards, 2024